Axios

HTTP 클라이언트

권기웅

1. 환경설정

1.1 환경설정

프로젝트 만들고 나서, yarn add axios

```
C:₩Windows₩system32₩cmd.exe - "node" "C:₩Users₩sundooedu₩AppData₩Roaming₩npm₩₩node_modules₩yarn₩bin₩yarn.js" start
D:₩Kwon₩wsReact>cd r09_axios

D:₩Kwon₩wsReact₩r09_axios>yarn add axios
yarn add v1.22.19
[1/4] Resolving packages...
[2/4] Fetching packages...
[3/4] Linking dependencies...
warning " > @testing-library/user-event@13.5.0" has unmet peer dependency "@testing-library/dom@>=7.21.4".
warning "react-scripts > eslint-config-react-app > eslint-plugin-flowtype@8.0.3" has unmet peer dependency
"@babel/plugin-syntax-flow@^7.14.5".
warning "react-scripts > eslint-config-react-app > eslint-plugin-flowtype@8.0.3" has unmet peer dependency
"@babel/plugin-transform-react-jsx@^7.14.9".
warning "react-scripts > react-dev-utils > fork-ts-checker-webpack-plugin@6.5.2" has unmet peer dependency
"typescript@>= 2.7".
warning "react-scripts > eslint-config-react-app > @typescript-eslint/eslint-plugin > tsutils@3.21.0" has u
nmet peer dependency "typescript@>=2.8.0 || >= 3.2.0-dev || >= 3.3.0-dev || >= 3.4.0-dev || >= 3.5.0-dev ||
 >= 3.6.0-dev || >= 3.6.0-beta || >= 3.7.0-dev || >= 3.7.0-beta".
[4/4] Building fresh packages...
success Saved lockfile.
success Saved 3 new dependencies.
info Direct dependencies
└─axios@1.1.3
info All dependencies
├─axios@1.1.3
├─form-data@4.0.0
└─proxy-from-env@1.1.0
Done in 4.67s.

D:₩Kwon₩wsReact₩r09_axios>yarn start
yarn run v1.22.19
```

2. Cross-Domain AJAX

2.1 Cross-Domain AJAX

weather.js

```
import axios from "axios";
import React, { useState } from "react";

const Weather = () => {
  const [w, setW] = useState(null);
  const getWeather = () => {
    axios
      .get(
        "https://api.openweathermap.org/data/2.5/weather?q=seoul&appid=baff8f3c6cbc28a4024e336599de28c4&units=metric&lang=kr"
      )
      .then((res) => {
        setW(res.data);
      });
  };
  return (
    <div>
      <h1>날씨 : {w !== null ? w.weather[0].description : "몰라"}</h1>
      <h2>기온 : {w !== null ? w.main.temp : "몰라"}℃</h2>
      <h2>습도 : {w !== null ? w.main.humidity : "몰라"}%</h2>
      <button onClick={getWeather}>날씨는</button>
    </div>
  );
};

export default Weather;
```

App.js

```
import "./App.css";
import Weather from "./weather";

function App() {
  return <Weather />;
}

export default App;
```

3. 요청-응답

3.1 GET 방식 요청

3.1.1 back-end

SpringMVC 프로젝트

pom.xml 에 추가

```
<dependency>
    <groupId>org.codehaus.jackson</groupId>
    <artifactId>jackson-mapper-asl</artifactId>
    <version>1.9.13</version>
</dependency>
```

Menu.java

```
public class Menu {
    private String m_name;
    private int m_price;

    public Menu() {
        // TODO Auto-generated constructor stub
    }

    public Menu(String m_name, int m_price) {
        super();
        this.m_name = m_name;
        this.m_price = m_price;
    }

    public String getM_name() {
        return m_name;
    }

    public void setM_name(String m_name) {
        this.m_name = m_name;
    }

    public int getM_price() {
        return m_price;
    }

    public void setM_price(int m_price) {
        this.m_price = m_price;
    }

}
```

MenuResult.java

```java
public class MenuResult {
    private String m_name;
    private String m_result;

    public MenuResult() {
        // TODO Auto-generated constructor stub
    }

    public MenuResult(String m_name, String m_result) {
        super();
        this.m_name = m_name;
        this.m_result = m_result;
    }

    public String getM_name() {
        return m_name;
    }

    public void setM_name(String m_name) {
        this.m_name = m_name;
    }

    public String getM_result() {
        return m_result;
    }

    public void setM_result(String m_result) {
        this.m_result = m_result;
    }

}
```

MenuDAO.java

```java
@Service
public class MenuDAO {

    public MenuResult reg(Menu m) {
        System.out.println(m.getM_name());
        System.out.println(m.getM_price());
        return new MenuResult(m.getM_name(), "등록 성공");
    }

}
```

MenuController.java

```java
@Controller
public class MenuController {

    @Autowired
    private MenuDAO mDAO;

    @RequestMapping(value = "/menu.reg", method = RequestMethod.GET, produces = "application/json; charset=utf-8")
    public @ResponseBody MenuResult menuReg(Menu m, HttpServletResponse res) {
        res.setHeader("Access-Control-Allow-Origin", "*");
        return mDAO.reg(m);
    }

}
```

3.1.2 front-end

menuRegForm.js

```
import axios from "axios";
import React, { useState } from "react";

const MenuRegForm = () => {
  const [name, setName] = useState("");
  const [price, setPrice] = useState("");

  const reg = () => {
    axios
      .get(
        "http://192.168.0.125/formmm/menu.reg?m_name=" + name + "&m_price=" + price
      )
      .then((res) => {
        alert(res.data.m_name+ " " + res.data.m_result);
        setName("");
        setPrice("");
      });
    alert("A");
  };

  return (
    <div>
      메뉴명 : <input value={name} onChange={(e) => { setName(e.target.value); }} />
      <br />
      가격 : <input value={price} onChange={(e) => { setPrice(e.target.value); }} />
      <br />
      <button onClick={reg}>등록</button>
    </div>
  );
};

export default MenuRegForm;
```

App.js

```
import "./App.css";
import MenuRegForm from "./menuRegForm";

function App() {
  return <MenuRegForm />;
}

export default App;
```

3.2 A를 안하려면

3.2.1 front-end

menuRegForm.js

```
import axios from "axios";
import React, { useState } from "react";

const MenuRegForm = () => {
  const [name, setName] = useState("");
  const [price, setPrice] = useState("");

  const reg = async () => {
    await axios
      .get(
        "http://192.168.0.125/formmm/menu.reg?m_name=" + name + "&m_price=" + price
      )
      .then((res) => {
        alert(res.data.m_name+ " " + res.data.m_result);
        setName("");
        setPrice("");
      });
    alert("끝");
  };

  return (
    <div>
      메뉴명 : <input value={name} onChange={(e) => { setName(e.target.value); }} />
      <br />
      가격 : <input value={price} onChange={(e) => { setPrice(e.target.value); }} />
      <br />
      <button onClick={reg}>등록</button>
    </div>
  );
};

export default MenuRegForm;
```

3.3 POST 방식 요청

3.3.1 back-end

web.xml 에 추가

```
<filter>
    <filter-name>encodingFilter</filter-name>
    <filter-class>org.springframework.web.filter.CharacterEncodingFilter</filter-class>
    <init-param>
        <param-name>encoding</param-name>
        <param-value>utf-8</param-value>
    </init-param>
</filter>
```

MenuController.java 에 추가

```
@RequestMapping(value = "/menu.reg.post", method = RequestMethod.POST, produces = "application/json; charset=utf-8")
public @ResponseBody MenuResult menuRegPost(Menu m, HttpServletResponse res) {
    res.setHeader("Access-Control-Allow-Origin", "*");
    return mDAO.reg(m);
}
```

3.3.2 front-end

menuRegFormPOST.js

```
import axios from "axios";
import React, { useState } from "react";

const MenuRegFormPOST = () => {
  const [m, setM] = useState({ m_name: "", m_price: "" });

  const reg = () => {
    axios
      .post("http://192.168.0.125/formmm/menu.reg.post/", null, {
        params: m,
      })
      .then((res) => {
        alert(res.data.m_name + " " + res.data.m_result);
        setM({ m_name: "", m_price: "" });
      });
  };
  const updateMenu = (e) => {
    setM({ ...m, [e.target.name]: e.target.value });
  };
  return (
    <div>
      메뉴명 : <input name="m_name" value={m.m_name} onChange={updateMenu} />
      <br />
      가격 : <input name="m_price" value={m.m_price} onChange={updateMenu} />
      <br />
      <button onClick={reg}>등록</button>
    </div>
  );
};

export default MenuRegFormPOST;
```

App.js

```
import "./App.css";
import MenuRegForm from "./menuRegForm";
import MenuRegFormPOST from "./menuRegFormPOST";

function App() {
  return (
    <>
      <MenuRegForm />
      <hr />
      <MenuRegFormPOST />
    </>
  );
}

export default App;
```

3.4 파일 업로드

3.4.1 back-end

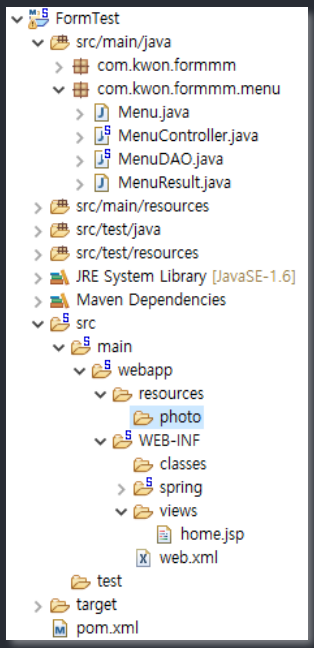
pom.xml 에 추가

```
<dependency>
    <groupId>com.jfinal</groupId>
    <artifactId>cos</artifactId>
    <version>26Dec2008</version>
</dependency>
```

Menu.java

```
public class Menu {
    private String m_name;
    private int m_price;
    private String m_photo;

    public Menu() {
        // TODO Auto-generated constructor stub
    }

    public Menu(String m_name, int m_price, String m_photo) {
        super();
        this.m_name = m_name;
        this.m_price = m_price;
        this.m_photo = m_photo;
    }

    public String getM_name() {
        return m_name;
    }

    public void setM_name(String m_name) {
        this.m_name = m_name;
    }

    public int getM_price() {
        return m_price;
    }

    public void setM_price(int m_price) {
        this.m_price = m_price;
    }
```

파일 업로드 될 폴더



MenuDAO.java 에 추가

```
public MenuResult reg2(Menu m, HttpServletRequest req) {
    try {
        String path = req.getSession().getServletContext().getRealPath("resources/photo");

        MultipartRequest mr = new MultipartRequest(req, path, 10485760, "utf-8", new DefaultFileRenamePolicy());

        m.setM_name(mr.getParameter("m_name"));
        m.setM_price(Integer.parseInt(mr.getParameter("m_price")));
        String m_photo = mr.getFilesystemName("m_photo");
        m_photo = URLEncoder.encode(m_photo, "utf-8");
        m_photo = m_photo.replace("+", " ");
        m.setM_photo(m_photo);

        System.out.println(path);
        System.out.println(m.getM_name());
        System.out.println(m.getM_price());
        System.out.println(m.getM_photo());

        return new MenuResult(m.getM_name() + "(" + m.getM_photo() + ")", "등록 성공");
    } catch (Exception e) {
        e.printStackTrace();
        return new MenuResult("등록", " 실패");
    }
}
```

MenuController.java 에 추가

```
@RequestMapping(value = "/menu.reg.file", method = RequestMethod.POST, produces = "application/json; charset=utf-8")
public @ResponseBody MenuResult menuRegFile(Menu m, HttpServletRequest req, HttpServletResponse res) {
    res.setHeader("Access-Control-Allow-Origin", "http://localhost:3000");
    res.setHeader("Access-Control-Allow-Credentials", "true");
    return mDAO.reg2(m, req);
}
```

3.4.2 front-end

menuRegFormFile.js

```
import axios from "axios";
import React, { useRef, useState } from "react";

const MenuRegFormFile = () => {
  const [name, setName] = useState(null);   const [price, setPrice] = useState(null);
  const [photo, setPhoto] = useState(null);
  const photoInput = useRef();

  const fd = new FormData();
  fd.append("m_name", name);    fd.append("m_price", price);
  fd.append("m_photo", photo);

  const reg = () => {
    axios
      .post("http://192.168.0.125/formmm/menu.reg.file", fd, {
        headers: { "Content-Type": "multipart/form-data" },
        withCredentials: true,
      })
      .then((res) => {
        alert(res.data.m_name+ " " + res.data.m_result);
        setName("");        setPrice("");
        photoInput.current.value = "";
      });
  };
  return (
    <div>
      메뉴명 : <input value={name} onChange={(e) => { setName(e.target.value); }} />
      <br />
      가격 : <input value={price} onChange={(e) => { setPrice(e.target.value); }} />
      <br />
      사진 : <input type="file" ref={photoInput} onChange={(e) => { setPhoto(e.target.files[0]); }} />
      <br />
      <button onClick={reg}>등록</button>
    </div>
  );
};

export default MenuRegFormFile;
```

App.js

```
import "./App.css";
import MenuRegForm from "./menuRegForm";
import MenuRegFormFile from "./menuRegFormFile";
import MenuRegFormPOST from "./menuRegFormPOST";

function App() {
  return (
    <>
      <MenuRegForm />
      <hr />
      <MenuRegFormPOST />
      <hr />
      <MenuRegFormFile />
    </>
  );
}

export default App;
```